

Alexiy Buynitsky

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EDUCATION

- UC San Diego** *La Jolla, CA* Sept 2025 – Jun 2027
Master of Science in Computer Science **GPA: 4.00/4.00**
Courses: Generative AI, Parallel Computing, Convex Optimization, Machine Learning Systems, Advanced Computer Vision, Search and Optimization
- Purdue University** *West Lafayette, IN* Aug 2022 – May 2025
Bachelors of Science in Computer Science, Bachelors of Science in Mathematics **GPA: 3.99/4.00**
Graduate Courses: Robot Learning, Robot Manipulation, Deep Learning, Compilers, Datamining & ML
Undergraduate Courses: Analysis of Algorithms, Complex Analysis, Abstract Algebra, Statistics, Physics E&M
Computer Architecture, Artificial Intelligence, Probability, Linear Algebra, Real Analysis
Systems Programming, Data Structures & Algorithms, C Programming, Discrete Math
- De Anza College** *Cupertino, CA* Jun 2021 – Jun 2022
Dual Enrolling HS Student **GPA: 4.00/4.00**
Courses: Differential Equations, Multivariable Calculus, C++ Programming, x86 Programming, Python Programming

EXPERIENCE

- Software Engineering Intern** | **Nvidia** | *Santa Clara, CA* June 2026 - Sept 2026
• Decision Making and Motion Planning for Autonomous Vehicles
- Graduate Researcher** | **UCSD Wang Lab** | *La Jolla, CA* Sept 2025 - Present
• Researching dexterous manipulation, tactile sensors, retargeting under the supervision of Prof. Xiaolong Wang
• Develop generalizable functional grasping policies for 12 & 6 DOF hands mounted on humanoids & manipulators
• Generate grasps via energy-based optimization in sim, performing Sim2Real transfer using segmented pointclouds
• Trained 3D Diffusion Policy from single real-world demonstration by augmenting trajectory segments in simulation using trained grasping policy, randomizing objects & object pose, and re-rendering scenes
- ML Engineering Intern** | **Persona AI** | *Houston, TX* May 2025 - Aug 2025
• Engineered architectural & training improvements to visuomotor policies / VLAs on frankas & mobile ALOHA
• Built teleoperation pipeline with Gello & SpaceMouse for large-scale data collection on bimanual manipulators
• Designed data processing pipeline, converting raw data into frequency-synchronized, model-agnostic datasets and generating model-specific datasets for training
• Created evaluation pipelines using digital twins in NVIDIA Isaac Lab and quantitative metrics
- Graduate Researcher** | **CoMMA Lab** | *West Lafayette, IN* Oct 2024 - May 2025
• Research Active Vision & Behavioral Cloning for manipulators under the supervision of Prof. Zachary Kingston
• Achieve 250× speedup in generating expert trajectories for point-to-point motion planning in tabletop environments using RRT-Connect with hardware-accelerated planning (VAMP)
• Extended M π Nets to learn point-to-point motion planning via Behavioral Cloning from expert trajectories generated by VAMP
- AI Engineering Intern** | **Armada AI** | *Remote* Oct 2023 - May 2025
• Develop spatially-aware Code-as-Policies methods for controlling PTZ Cameras on the edge
• Perform model distillation by generating synthetic data and finetuning LLMs using SFT and DPO
• Developed VLM/LLM VideoQA agent for real-time video Q&A for security camera footage
- Undergraduate Researcher** | **CoRAL Lab** | *West Lafayette, IN* Aug 2023 - Sept 2024
• Conduct research on robotic learning under the supervision of Prof. Ahmed Qureshi
• Extended Unitree simulator to support Unitree B1 Quadruped Robot in Gazebo and PyBullet
• Advance Motion Planning in dynamic environments via Network Time Fields and Sign Distance Fields
• Teach robots to navigate through Purdue with custom knowledge using LLMs, RAG, and vector databases
- Engineering Intern** | **SpaceX** | *Redmond, WA* May 2023 - Aug 2023
• Develop mechatronic / software solutions for quicker manufacturing and assembly of Starlink Satellites
• Prototype satellite assembly cells, working with 6-axis robotics arms, CV, actuators, sensors, & safety hardware
• Achieve 80x speedup between PLC & CV software by developing an IP-style communication library
- Undergraduate ML Researcher** | **Duality Lab x Google** | *West Lafayette, IN* Jan 2023 - May 2023
• Build data pipeline for MaskFormer and Mask2Former for TensorFlow 2's Model Garden
• Generate, and load TFRecords for panoptic segmentation, adding data augmentations, configs and dataloaders

PUBLICATIONS AND PATENTS

Liu, Y., **Buynitsky, A.**, Ni, R., and Qureshi, A.H., 2025. Physics-informed Neural Motion Planning via Domain Decomposition in Large Environments. Published in *2025 IEEE International Conference on Intelligent Robots & Systems (IROS)*

Chen, W., Liu, Y., **Buynitsky, A.**, and Qureshi, A.H., 2025. Self-supervised Hierarchical Robot Navigation Policy Learning using Physics Priors in Unknown Environments. Published in *2025 IEEE International Conference on Intelligent Robots & Systems (IROS)*

Buynitsky, A., and Kingston, Z., 2025. Faster Behavioral Cloning with Hardware-Accelerated Motion Planning. Published in RoboARCH Workshop at *2025 IEEE International Conference on Robotics & Automation (ICRA)*

Buynitsky, A., Ehsani, S. and Mishra, P.K., Armada Systems Inc, 2025. *Robotic Control Using Natural Language Commands* U.S. Patent 12289517 B1.

Buynitsky, A., Ehsani, S. and Mishra, P.K., 2024. Camera Control at the Edge with Language Models for Scene Understanding. Published in *11th International Conference on Control, Automation and Robotics (ICCAR)*

PROJECTS

3D Consistency in Video Generation | *Pytorch, Python* Sept 2025 – Dec 2025

- Introduce metric based on 6D Chamfer Distance to quantify rigid 3D consistency in video generation models
- Build pipeline to extract point clouds from video frames, performing 3D registration across adjacent point clouds

Edge Probing for Decoder-only Transformers | *Pytorch, Python, NLP* Mar 2025 – June 2025

- Developed a two-stage interpretability framework (Block Pruning and Edge Probing) to identify which transformer blocks contain NLP-Related Information on 7 NLP tasks (POS, NER, CP, SR, UD, SRL, CR), achieving up to 92% sparsity in hidden block embeddings while maintaining comparable performance on NLP tasks
- Finetune GPT-2 on auxiliary NLP objectives, reducing perplexity by 9% compared to finetuning only on self-supervised objective using OWT

Gesture Controlled HCI | *Pytorch, Flask, MongoDB* Jan 2024 – Mar 2024

- Built a continuous learning model to detect hand poses at 30FPS allowing for customizable hand poses
- Categorized hand gestures through VLMs and vector databases and create custom actions using open-interpreter

Robotics Mini-Projects | *Pytorch, Gazebo, PyBullet, ROS* Jan 2024 – May 2024

- Implement RRTConnect, RRT* for cars and 6-DOF arms; Iterative/Analytic PID for Quadruped robots and 2-DOF arms; MPNet in 2D/3D environments; VPG for 2-DOF arm

1st Place Purdue BoilerMake X Hackathon Dagshub | *Pytorch, MLFlow, DVC, Dagshub* Jan 2023

- Used seq2seq model to study key factors affecting air quality. Created a robust, modular testing environment for time-series forecasting with any data through MLFlow, DVC, and git using Dagshub

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Twincat3, C#, Bash, x86 Assembly, SQL

Frameworks: Pytorch, ROS, CUDA, SIMD, MPI, Tensorflow

Platforms/Tools: Docker, Conda, Linux, VIM, Git/Github, Dagshub, Onshape